Markscheme

Mathematics: analysis and approaches

Practice question bank

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14. (a)

Consider the function $f(x) = \dfrac{3x+4}{(x-2)(x+3)}$.

Write $f(x)$ as the sum of partial fractions in the form $\dfrac{A}{x-2} + \dfrac{B}{x+3}$.

$$3x + 4 = A(x+3) + B(x-2)$$ *(M1)*

substituting $x = 2$: $10 = 5A \Rightarrow A = 2$ **A1**

substituting $x = -3$: $-5 = -5B \Rightarrow B = 1$ **A1**

[3 marks]